

TARGU MURES, Romania

WMO: 151450

Lat: 46.5281N

Lon: 24.5227E

Elev: 311

StdP: 97.64

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-12.8	-17.8	0.8	-15.0	-14.4	1.1	-12.0	9.2	-0.3	7.4	0.5	1.0	90	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	32.2	20.7	30.9	20.1	29.1	19.5	22.1	29.2	21.2	28.2	20.3	27.0	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.3	24.7	19.0	14.3	23.6	18.1	13.5	22.6	66.5	29.4	62.9	28.3	59.7	27.1	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.1	4.9	DB	-19.6	35.1	3.7	1.8	-22.2	36.4	-24.4	37.5	-26.5	38.5	-29.2	39.8	(4)
(5)			WB	-19.7	23.5	3.7	1.3	-22.4	24.5	-24.6	25.3	-26.6	26.0	-29.3	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.4	0.4	5.1	10.9	15.7	19.2	20.9	20.7	15.8	10.2	4.5	-1.0	(6)	
	DBStd	9.13	4.99	4.98	4.37	3.66	3.30	3.02	2.83	2.87	3.60	4.08	4.59	4.75	(7)	
	HDD10.0	1411	384	269	158	32	2	0	0	0	2	48	173	342	(8)	
	HDD18.3	3274	642	502	411	223	93	26	9	11	89	251	416	600	(9)	
	CDD10.0	1429	0	0	6	59	178	275	338	333	177	56	7	0	(10)	
	CDD18.3	252	0	0	0	1	11	51	89	85	15	1	0	0	(11)	
	CDH23.3	2928	0	0	1	29	200	562	943	939	239	16	0	0	(12)	
	CDH26.7	1001	0	0	0	3	34	171	357	368	66	2	0	0	(13)	
(14) Wind	WSAvg	1.9	1.6	1.8	2.3	2.3	2.1	2.0	1.9	1.8	1.8	1.6	1.6	1.5	(14)	
(15) Precipitation	PrecAvg	604	28	25	32	50	70	92	77	55	55	48	37	38	(15)	
	PrecMax	787	62	61	75	91	153	193	125	109	131	93	74	86	(16)	
	PrecMin	438	10	7	7	5	23	26	27	16	5	3	0	9	(17)	
	PrecStd	95	14	13	18	25	28	38	30	27	30	25	16	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	15.0	20.9	26.8	29.8	32.8	34.9	35.0	31.2	25.1	19.8	11.2	(19)	
		MCWB	7.3	9.4	11.9	15.3	18.3	21.2	20.6	20.6	18.4	15.9	13.7	8.3	(20)	
	2%	DB	7.0	11.8	17.9	23.2	27.3	30.8	32.2	32.6	28.3	22.9	16.2	8.1	(21)	
		MCWB	5.5	7.8	10.4	13.8	17.2	20.5	21.2	20.4	17.7	15.0	11.8	6.7	(22)	
	5%	DB	5.0	9.0	15.0	20.9	25.8	28.9	30.7	30.8	26.0	20.1	13.7	6.1	(23)	
		MCWB	3.9	6.1	9.2	12.7	16.9	19.9	20.4	20.0	16.9	14.2	10.3	5.3	(24)	
	10%	DB	3.1	6.8	12.8	18.9	23.6	26.9	28.6	28.8	23.8	17.6	11.0	4.7	(25)	
		MCWB	2.4	4.8	8.0	11.8	16.0	19.0	19.7	19.4	16.4	12.9	8.8	4.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	9.9	12.9	16.2	20.6	23.1	23.5	23.4	20.2	17.3	14.2	8.9	(27)	
		MCDB	9.6	14.1	18.9	24.1	26.4	29.9	30.7	30.3	28.3	23.1	18.2	10.4	(28)	
	2%	WB	5.6	8.1	11.1	14.8	18.7	21.7	22.1	21.9	18.6	15.9	12.2	6.7	(29)	
		MCDB	6.5	11.1	16.4	21.6	24.8	28.3	29.3	29.0	25.5	20.9	15.4	7.8	(30)	
	5%	WB	3.9	6.5	9.7	13.6	17.4	20.5	21.2	20.9	17.7	14.7	10.7	5.3	(31)	
		MCDB	4.7	8.5	14.3	19.2	23.7	26.8	28.2	28.1	23.9	19.1	13.3	6.0	(32)	
	10%	WB	2.4	4.8	8.3	12.5	16.5	19.5	20.3	19.9	16.8	13.5	9.0	3.9	(33)	
		MCDB	3.0	6.5	12.1	17.4	22.2	25.4	26.8	26.7	22.3	17.1	10.9	4.4	(34)	
(35) Mean Daily Temperature Range	MDBR		6.9	8.9	11.3	12.6	12.7	12.6	13.1	13.8	12.7	12.2	9.0	6.1	(35)	
	5% DB	MCDBR	8.4	12.1	16.3	16.8	16.6	15.9	16.6	17.3	17.2	16.4	13.5	8.4	(36)	
		MCWBR	6.6	8.2	9.4	8.4	7.7	7.0	6.5	6.4	7.3	9.2	9.0	6.7	(37)	
	5% WB	MCDBR	8.0	11.5	14.5	14.5	14.5	14.4	14.8	15.1	15.0	14.6	12.1	8.1	(38)	
		MCWBR	6.5	8.1	8.8	7.7	7.4	6.9	6.4	6.3	7.0	8.4	8.3	6.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.313	0.341	0.382	0.434	0.416	0.413	0.433	0.438	0.398	0.363	0.340	0.315	(40)	
	taud		2.331	2.291	2.240	2.123	2.238	2.280	2.229	2.211	2.292	2.377	2.389	2.359	(41)	
	Ebn at Noon		769	819	835	820	851	854	830	808	809	785	732	722	(42)	
	Edh at Noon		76	97	118	145	134	130	135	132	111	88	72	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.29	2.15	3.26	4.36	5.40	5.84	5.78	5.32	3.78	2.59	1.46	0.98	(44)	
	RadStd		0.17	0.23	0.38	0.43	0.49	0.49	0.48	0.47	0.43	0.32	0.19	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	+1.09	N/A	N/A	N/A	-167	+95	+54
(47) Regional (72 neighbors)		+0.72	N/A	N/A	+0.94	N/A	N/A	-155	-220	+87	+18

Nomenclature: See separate page